

Generated 2026-10-06 from branch `claude/practical-hamilton-sspq2e`, file `pvv_score/results/REPORT.md`. Ranking as of the 2026-10-05 close.

PVV Score: price / volume / volatility selection for 21-63 day forward risk-adjusted returns

Universe: 503 Finviz S&P 500 constituents (file `data/universe_finviz.csv`), sector mapped to SPDR ETF. Data: yfinance daily OHLCV, dividend/split adjusted, 2012-2026-10-05. Backtest window 2014-01-01 onward; **the 2022-10 onward window ('recent') is the primary evidence** per the mandate, the full window is the secondary check.

All statistics are out-of-sample where labelled OOS. Nothing here uses fundamentals, earnings or valuation.

1. Bottom line

- Two readings carry out-of-sample signal for 21-63d forward Sharpe in this universe in the current regime: relative-strength leadership (`rs_lead_126` : RS line nearer its 6-month high than price is to its own) and 12-1 momentum. Both survive beta-neutralisation (recent-window 42d IC 0.03-0.05, NW t 2.1-2.4). Both were dead 2014-2021.
- The setup that works depends on beta. High-beta names: buy persistence (momentum, trailing Sharpe, proximity to highs, recent 2x-off-low crossings; the 100%-off-low ceiling is the wrong sign there). Low-beta names: price-structure factors die and quiet volume-led accumulation takes over (volume dry-up vs 250d, falling dollar volume, OBV leading price, shallow drawdowns, low idiosyncratic vol, and the dry-up x tight-range x near-high interaction).
- The live ranking (`current_rankings.csv`) is the within-beta-bucket walk-forward composite built from those weights: recent-window OOS 42d IC 0.047, 63d 0.065, top-vs-bottom decile 42d Sharpe spread +0.29 annualised, 57% of windows positive. A pooled (beta-blind) composite has no edge.
- Compression / NR7 / Bollinger squeeze, ignition-day volume multiples, accumulation-day counts, gap behaviour, streaks and close-location factors showed no 1-3 month signal in S&P 500 names, pooled. The LightGBM case-study model, in five configurations, did not beat the two single factors out of sample (recent 42d IC 0.00-0.02) and is reported but not used in the ranking.

2. Method

- **Targets.** For every (date, stock): realised Sharpe and Sortino of the stock over $t+1..t+h$ ($h = 21, 42, 63$) minus the same for SPY and for the stock's sector ETF (XLRE/XLC spliced to XLF/XLK before inception). Also the information-ratio form (Sharpe of stock-minus-SPY). The ML label is the per-date percentile of SPY-excess Sharpe averaged over the three horizons.
- **Eligibility (point-in-time).** Price $\geq \$5$, 63d median dollar volume $\geq \$10M$, ≥ 252 sessions of history. Survivorship caveat: the universe is today's constituents, so absolute levels are biased up; cross-sectional ranks are far less affected.
- **Factors.** 86 point-in-time factors in five families (price structure, volume, volatility, relative strength, run character) plus two user-described interaction setups. Each verified against hand calculations and a perturbation test for look-ahead (none).
- **Statistics.** Daily cross-sectional Spearman IC; Newey-West t-stats with lag = horizon because forward windows overlap; Benjamini-Hochberg across the 86-factor zoo; decile spreads in raw excess-Sharpe units; 21-day rank autocorrelation (turnover).
- **Conditioning.** Everything re-run inside trailing-beta terciles, 20d realised-vol terciles, sector groups (defensive / cyclical / growth), per sector, and with sector-neutral ranks.
- **Case-study model.** LightGBM trained walk-forward by calendar year (2017-2026), 63-day embargo, 2-year half-life time-decay weights, every 5th day sampled for training. Four pre-specified regularised variants were also run. Nearest-neighbour analogs (50 historical cases, max 3 per ticker) give each live name its realised-outcome evidence.
- **Composite.** Factor selection and weights re-estimated each year on training data only ($|NW\ t| \geq 1.5$ on the trailing 3y, sign agreement with full history, greedy de-duplication at $|\rho| > 0.7$, weights = recency-blended IC). Pooled and beta-bucketed variants.

3. Single-factor results (SPY-excess Sharpe, mean IC across 21/42/63d)

Power note: the recent window is $\sim 1,000$ sessions = roughly 24 independent 42-day windows, so $|t|$ around 2 is the attainable ceiling for a genuine IC of 0.05-0.07. **No factor survives Benjamini-Hochberg at 5% in either window.** The evidence is in coherent clusters, not individual rows.

	group	prior	ic_spy_full	t_spy_full	ic_spy_recent	t_spy_recent	ic_sec_recent	t_sec_recent	sign_stable	ic42_pos_years	d10_d1_spread_recent_42	autocorr_21
rs_lead_126	relative	1	-0.002	-0.127	0.066	2.436	0.013	0.705	False	0.538	0.515	0.72
ceiling_2x_low	run	-1	0.01	1.151	0.038	2.275	0.019	1.662	True	0.615	0.28	0.743
corr_spy_63	relative	-1	0.02	1.176	0.072	2.119	-0.001	-0.1	True	0.615	0.437	0.848
parkinson_cc_20	vol	0	-0.004	-0.483	-0.029	-1.977	0.008	1.081	True	0.462	-0.182	0.081
dist_52w_low	run	0	0.002	0.154	0.047	1.915	0.027	1.569	True	0.538	0.417	0.869

	group	prior	ic_spy_full	t_spy_full	ic_spy_recent	t_spy_recent	ic_sec_recent	t_sec_recent	sign_stable	ic42_pos_years	d10_d1_spread_recent_42	autocorr_21
up_capture_63	relative	1	0.005	0.219	0.069	1.911	0.001	0.024	True	0.538	0.55	0.817
overnight_intraday_vol_20	vol	0	0.011	1.099	0.034	1.908	-0.001	-0.083	True	0.692	0.211	0.203
beta_63	relative	0	-0.004	-0.23	0.072	1.817	-0.007	-0.403	False	0.538	0.539	0.88
days_since_2x_low	run	0	-0	-0.013	-0.037	-1.791	-0.028	-2.47	True	0.385	-0.262	0.923
gap_share_21	price	0	0.014	1.335	0.037	1.765	0.007	0.623	True	0.692	0.32	0.26
beta_252	relative	0	0.002	0.06	0.072	1.727	-0.013	-0.661	True	0.462	0.398	0.985
mom_12_1	price	1	0.005	0.374	0.04	1.66	0.043	2.288	True	0.615	0.277	0.892
up_gap_freq_21	price	1	-0.009	-0.532	0.041	1.583	-0.003	-0.188	False	0.538	0.318	0.452
down_capture_63	relative	-1	0.003	0.112	0.051	1.461	-0.02	-1.236	True	0.538	0.308	0.792
dn_vol_63	vol	-1	-0.011	-0.681	0.043	1.457	-0.014	-0.846	False	0.615	0.239	0.845
rv20	vol	0	-0.015	-0.961	0.04	1.37	-0.01	-0.67	False	0.538	0.274	0.545
rs_newhigh_price_not_20	relative	1	-0.006	-0.694	-0.017	-1.362	0	0.046	True	0.692	0.011	0.196
dvol_trend_21_126	volume	0	-0.002	-0.333	-0.015	-1.343	0.011	1.395	True	0.462	-0.087	0.242
accum_dist_50	volume	1	-0.009	-1.034	-0.019	-1.255	0.006	0.506	True	0.769	-0.173	0.563
rv_ratio_5_20	vol	0	0.001	0.384	0.006	1.188	0.002	0.671	True	0.538	0.026	-0.006
base_tightness_63	price	-1	-0.011	-0.734	0.033	1.136	-0.004	-0.283	False	0.615	0.21	0.752
bbw_squeeze_126	price	-1	-0.004	-0.64	0.015	1.134	0.011	1.316	False	0.538	0.121	0.108
dn_up_vol_asym_63	vol	-1	0.011	1.059	0.016	1.087	-0.001	-0.058	True	0.538	0.059	0.614
rs_sec_21	relative	1	0.002	0.23	0.014	1.087	0.008	0.603	True	0.462	0.237	-0.01
obv_price_div_63	volume	1	0.003	0.454	0.011	1.08	0.007	0.814	True	0.615	0.077	0.575

`d10_d1_spread_recent_42` is the annualised 42d Sharpe difference between the factor's top and bottom decile (recent window). `autocorr_21` near 1 = slow factor; near 0 = daily-turnover factor.

Reading of the clusters

- **Trend-persistence / leadership cluster** (`rs_lead_126` , `mom_12_1` , `dist_52w_low` , `up_capture_63` , `days_since_2x_low` negative, `ceiling_2x_low` positive): the only group with a consistent positive sign in the recent window and a positive sector-excess reading. Note `ceiling_2x_low` comes in with the OPPOSITE sign to the prior: names already 100% off their 52w low have continued to produce better forward Sharpe in 2022-2026. In the pooled sample the "ceiling" is not a ceiling.
- **Beta / correlation cluster** (`beta_63` , `beta_252` , `corr_spy_63` , `rv20` , `dn_vol_63`): positive in the recent window, flat over the full window, negative vs sector. This is the bull-regime beta bet, not stock selection. The composite and the model are therefore also evaluated beta-neutral (section 5).
- **Compression / dry-up setups** (`range_comp_*` , `bbw_*` , `nr7_count_20` , `vol_dry_*` , `dryup_x_tight`): no pooled signal at these horizons. They show up only conditionally (section

4), in low-vol / defensive names.

- **Ignition / volume-burst factors** (`ignition_mult_10` , `vol_z_1` , `big_vol_updays_net_21` , `ignite_x_rs`): no signal for 21-63d forward Sharpe in S&P 500 names. These are shorter-horizon or small-cap phenomena.
- **Short-term price factors** (`ret_5` , `clv_5` , `streak_now` , `pos_in_20d_range`): nothing. Daily-turnover factors with no 1-3 month payoff.

4. Conditional results: where each factor works (recent window, 42d, NW t-stats)

	beta_low	beta_mid	beta_high	vol_low	vol_mid	vol_high	grp_defensive	grp_cyclical	grp_growth	sector_neutral_vs_sector
mom_12_1	1.41	1.09	3.59	2.06	2.54	2.02	0.99	1.72	0.82	1.79
obv_price_div_63	0.5	-0.34	0.91	0.36	0.91	-0.72	3.56	-0.67	-0.8	0.32
dvol_trend_21_126	-2.93	-0.3	1.47	-1.76	-1.23	-0.49	-1.3	-1.36	1.16	0.2
skew_63	0.44	0.33	-2.1	-0.16	-0.92	1.02	-2.24	0.56	2.88	0.45
corr_spy_63	0.83	2.72	1.17	2.09	1.83	1.85	0.72	1.83	0.43	1.72
ceiling_2x_low	1.61	-1.1	2.71	0.76	2.17	2.22	-0.02	1.02	2.58	2.2
idio_vol_63	-1.1	-2.39	0.47	-2.12	-2.69	1.03	-0.04	-1.57	1.75	-0.06
days_since_2x_low	-0.1	0.79	-2.68	-1.08	-1.62	-1.51	0.13	-0.82	-2.25	-2.54
rs_lead_126	1.55	1.97	1.77	2.59	1.68	2.35	1.31	2.43	1.18	2.11
capture_spread_63	-0.93	1.63	1.7	-0.41	0.22	1.39	-1.64	-0.07	2.54	0.36
days_since_52w_high	-1.11	-0.59	-2.52	-0.53	-1.43	-1.55	-0.42	-0.63	-0.6	-0.75
dist_52w_low	0.85	1.59	2.5	1.9	2.11	2.01	0.31	1.48	1.82	1.79
up_day_frac_63	-0.94	0.87	2.44	0.23	0.14	0.27	-0.11	-1.09	0.76	-0.1
dn_up_vol_asym_63	0.72	-1.12	0.5	0.5	0.98	-0.91	2.33	1.07	-2.42	0.57
base_depth_126	-0.14	2.36	1.1	0.42	0.85	0.36	-0.86	0.34	0.85	0.28
rs_sec_21	1.19	2.35	1.74	1.77	1.85	2	0.43	0.71	1.71	0.91
rs_sec_63	-0.33	2.34	1.33	0.75	0.24	1.28	-0.84	-0.76	1.47	-0.4
rv_ratio_20_60	1.48	-0.72	-0.86	0.33	-0.16	-2.3	0.94	0.63	-1.08	-0.07
rs_dist_high_126	-0.17	2.3	1.18	1.57	1.1	0.65	-0.88	1.16	0.98	0.75
base_depth_63	-0.55	2.14	0.91	0.04	0.44	0.29	-1.14	-0.13	0.85	-0.17
vol_dry_20_250	-1.41	-0.84	0.81	-2.12	-1.23	-0.52	-0.32	0.07	0.17	0.21
days_since_20pct_dd	0.83	1.64	2.09	0.31	1.31	0.6	-0.17	0.47	-0.18	0.32
dd_from_252_high	0.32	1.71	2.06	0.67	1.49	0.9	-0.57	0.74	1.02	0.62
ignition_mult_10	1.39	-1.25	-0.16	0.49	-0.27	1.21	1.18	-0.35	2.01	1.62
rs_newhigh_price_not_20	-0.07	-0.61	0.07	0.22	-0.67	-0.65	-2.01	-0.28	-0.3	-0.23
gap_share_21	1.48	0.92	0.83	2	1.45	1.07	0.51	1.01	1.29	1.46
dist_52w_high	0.34	1.71	2	0.64	1.46	0.86	-0.54	0.63	0.96	0.57
beta_63	0.32	1.5	1.24	1.77	1.77	1.99	0.62	1.24	1.3	1.59
rv_ratio_5_20	1.66	0.24	0.64	1.08	1.87	1.98	0.87	1.15	-0.08	0.42

	beta_low	beta_mid	beta_high	vol_low	vol_mid	vol_high	grp_defensive	grp_cyclical	grp_growth	sector_neutral_vs_sector
up_day_frac_21	-0.76	1.48	1.98	0.71	0.99	0.51	-0.08	0.34	0.97	0.81

Multiple-testing note: 86 factors x 10 buckets = 860 tests; the expected maximum $|t|$ under the null is about 3.2, so single cells near 3 are not individually reliable. What is reliable is the pattern:

- **High beta / growth:** the trend-persistence cluster is strongest here (`mom_12_1` $t=3.6$ in the high-beta tercile, `days_since_52w_high` -2.5, `ceiling_2x_low` +2.7, `days_since_2x_low` -2.7). In high-beta names, buy strength, recent 2x-off-low crossings, and proximity to highs.
- **Low beta / low vol / defensive:** the sign of several price factors flips or dies, and volume-structure factors appear instead: `obv_price_div_63` +3.6 in defensives (volume leading price), `vol_dry_20_250` -2.1 in the low-vol tercile (dry-up works for utilities/staples-type names), `dvol_trend_21_126` -2.9 in low beta (quiet names outperform), `idio_vol_63` negative in low/mid beta. This is the DUK-vs-NVDA asymmetry the mandate anticipated, and it is why the composite is fitted per beta bucket.
- **Per sector** (`conditional_ic_42_per_sector.csv`): indicative only (18-80 names per date). Notable: Utilities `rs_lead_126` +3.1 and dry-up negative; Consumer Defensive `ceiling_2x_low` -3.2 (the ceiling DOES hold in staples) and compression factors positive; Financials show reversal (`ret_63` -2.3).

5. Out-of-sample predictive models (walk-forward 2017-2026)

5a. Case-study model (LightGBM) vs single-factor baselines

lightgbm_walkforward

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.01	1.094	0.012	1.624	0.01	1.38
(21, 'recent')	-0	-0.015	0.004	0.349	0.017	1.72
(42, 'full')	0.013	1.066	0.014	1.56	0.014	1.624
(42, 'recent')	0.003	0.198	0.003	0.204	0.013	1.071
(63, 'full')	0.018	1.21	0.017	1.844	0.016	1.641
(63, 'recent')	0.018	0.884	0.009	0.788	0.009	0.613

baseline_mom_12_1

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.018	1.347	0.026	2.286	0.013	1.325
(21, 'recent')	0.038	1.918	0.044	2.534	0.037	2.337

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(42, 'full')	0.015	0.847	0.027	1.879	0.008	0.639
(42, 'recent')	0.041	1.598	0.048	2.361	0.043	2.097
(63, 'full')	0.014	0.659	0.03	1.93	0.006	0.414
(63, 'recent')	0.041	1.464	0.054	2.609	0.051	2.429

baseline_rs_lead_126

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.009	0.718	0.006	0.64	-0.001	-0.085
(21, 'recent')	0.048	2.261	0.026	2.058	0.005	0.448
(42, 'full')	0.015	0.897	0.006	0.524	-0	-0.007
(42, 'recent')	0.068	2.393	0.034	2.057	0.015	0.853
(63, 'full')	0.017	0.92	0.007	0.548	-0.001	-0.079
(63, 'recent')	0.082	2.654	0.041	2.237	0.017	0.814

baseline_beta_252

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.008	0.415	0.001	0.508	-0.019	-2.127
(21, 'recent')	0.05	1.538	-0.004	-1.427	-0.013	-0.837
(42, 'full')	0.02	0.775	0.004	1.248	-0.015	-1.164
(42, 'recent')	0.076	1.734	-0.003	-0.859	-0.011	-0.537
(63, 'full')	0.027	0.868	0.003	0.978	-0.014	-0.896
(63, 'recent')	0.089	1.909	-0.006	-1.307	-0.014	-0.608

LightGBM 42d IC by year: 2017: -0.011, 2018: +0.037, 2019: +0.141, 2020: +0.006, 2021: -0.005, 2022: -0.042, 2023: -0.037, 2024: +0.006, 2025: +0.047, 2026: -0.021

Regularised variants (rank-only features, 7 leaves, 300 rounds; ridge; with/without decay; no market-context features):

gbm_reg_rankonly_decay

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.007	0.661	0.005	0.662	0.011	1.688
(21, 'recent')	0.006	0.385	0	0.023	0.007	0.938
(42, 'full')	0.011	0.823	0.009	0.915	0.016	1.824
(42, 'recent')	0.016	0.752	0.001	0.095	0.01	1.139
(63, 'full')	0.013	0.817	0.012	1.226	0.02	1.874
(63, 'recent')	0.023	0.935	0.007	0.504	0.009	0.93

gbm_reg_rankonly_nodecay

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.003	0.276	0.003	0.409	0.01	1.489
(21, 'recent')	-0.013	-0.905	-0.011	-1.047	-0.002	-0.314
(42, 'full')	0.008	0.611	0.008	0.821	0.016	1.805
(42, 'recent')	-0.002	-0.105	-0.009	-0.722	-0.001	-0.094
(63, 'full')	0.015	0.929	0.014	1.335	0.024	2.122
(63, 'recent')	0.013	0.669	0.001	0.083	0.003	0.358

ridge_rank_decay

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.011	0.95	0.009	0.945	0.016	2.414
(21, 'recent')	0.011	0.564	0.002	0.152	0.012	1.539
(42, 'full')	0.014	0.911	0.007	0.65	0.014	1.479
(42, 'recent')	0.023	0.927	0	0.025	0.005	0.533
(63, 'full')	0.016	0.922	0.009	0.789	0.016	1.283
(63, 'recent')	0.032	1.149	0.004	0.247	-0.003	-0.264

gbm_reg_noctx_decay

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.007	0.667	0.008	0.95	0.004	0.665
(21, 'recent')	-0	-0.001	-0	-0.011	0.009	1.233
(42, 'full')	0.01	0.732	0.009	0.901	0.009	1.003
(42, 'recent')	0.01	0.468	0	0.008	0.012	1.168
(63, 'full')	0.014	0.846	0.012	1.12	0.012	1.149
(63, 'recent')	0.022	0.941	0.006	0.41	0.013	1.124

5b. Walk-forward linear composite

composite_pooled_wf

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.005	0.37	-0.001	-0.058	0.008	1.115
(21, 'recent')	-0.014	-0.79	-0.027	-2.276	-0.013	-1.288
(42, 'full')	0.006	0.376	-0.002	-0.124	0.006	0.561
(42, 'recent')	-0.005	-0.231	-0.035	-2.844	-0.019	-1.65
(63, 'full')	0.01	0.526	0.002	0.138	0.007	0.49
(63, 'recent')	0.015	0.648	-0.024	-1.685	-0.02	-1.837

composite_betabucket_wf

	IC vs SPY	t	IC beta-neutral	t_bn	IC vs sector	t_sec
(21, 'full')	0.009	0.722	0.006	0.812	-0.005	-0.716
(21, 'recent')	0.027	1.109	0.002	0.205	0.002	0.163
(42, 'full')	0.018	1.086	0.006	0.699	-0.003	-0.361
(42, 'recent')	0.047	1.463	0.006	0.471	0.005	0.268
(63, 'full')	0.025	1.346	0.008	0.827	-0.002	-0.141
(63, 'recent')	0.065	1.905	0.015	1.097	0.006	0.294

6. Live scoring

`current_rankings.csv` columns: `final_score` = within-beta-bucket composite percentile (the ML percentile is reported alongside, not blended, because of its OOS record);

`analog_med_xsSharpe42` / `analog_pct` / `analog_hit_rate` = realised 42d SPY-excess Sharpe of the 50 nearest historical cases (median, its percentile across today's names, share > 0); key factor readings follow.

Final composite weights by beta bucket

	low	mid	high
accum_dist_50	nan	0.033	nan
base_depth_126	nan	0.091	nan
capture_spread_63	nan	nan	0.075
ceiling_2x_low	nan	nan	0.081
clv_21	nan	0.042	nan
clv_5	nan	0.02	nan
comp_then_exp	-0.036	nan	nan
corr_spy_63	nan	0.087	nan
days_since_20pct_dd	0.248	0.075	0.082
days_since_2x_low	nan	nan	-0.079
days_since_52w_high	nan	nan	-0.093
dist_52w_low	nan	0.061	nan
dryup_x_tight	0.16	0.055	nan
dvol_trend_21_126	-0.101	nan	0.041
higher_low_cadence	nan	0.046	nan
idio_vol_63	-0.242	-0.076	nan
max_up_streak_21	nan	nan	0.032
mdd_63	0.213	nan	nan
mom_12_1	nan	nan	0.136
mom_6_1	nan	0.039	nan
pullback_21	nan	0.053	0.029

		low	mid	high
pullback_duration_21		nan	-0.037	nan
rs_lead_126		nan	0.052	0.056
rs_sec_21		nan	0.054	nan
rs_sec_63		nan	0.065	nan
sharpe_126		nan	nan	0.095
skew_63		nan	nan	-0.039
up_day_frac_21		nan	0.056	0.052
up_day_frac_63		nan	0.059	0.075
vol_cv_20		nan	nan	0.036

Top 30 as of 2026-10-05

ticker	sector	beta_bucket	final_score	ml_score	comp_score	analog_pct	analog_hit_rate	mom_12_1	dist_52w_high	dist_52w_low	rs_lead_126	beta_252	rv20_pct_252	vol_dry_20_250	obv_price_div_63
BRK-B	Financial	low	1	0.53	0.51	0.87	0.48	0.02	-0.06	0.09	-0.06	0.01	0.86	0.99	1.36
NTAP	Technology	high	1	0.56	0.66	0.32	0.36	0.57	-0.04	1.41	-0.01	1.31	1.3	1.02	1.3
EXPD	Industrials	mid	1	0.5	0.77	0.58	0.34	0.54	-0.01	0.74	-0.02	0.53	1.05	0.87	1.13
ATO	Utilities	low	0.99	0.58	0.49	0.32	0.44	0.02	-0.17	0.02	-0.11	-0.16	0.68	0.9	-0.09
NDSN	Industrials	mid	0.99	0.54	0.75	0.96	0.54	0.38	-0.01	0.52	-0.02	0.84	0.87	0.92	-0.59
P	Technology	high	0.99	0.63	0.64	0.42	0.36	0.11	-0	1.53	0	2.44	0.98	2.09	0.7
ED	Utilities	low	0.99	0.63	0.47	0.47	0.38	0.15	-0.1	0.12	-0.14	-0.46	0.79	1	1.01
IEX	Industrials	mid	0.99	0.52	0.71	0.13	0.3	0.37	-0.03	0.5	-0.02	0.65	0.71	0.88	1.12
CRWD	Technology	high	0.99	0.56	0.61	0.4	0.38	0.73	-0.01	2.18	0	1.49	1.11	0.87	-1
REG	Real Estate	low	0.98	0.57	0.45	0.36	0.4	0.09	-0.14	0.11	-0.08	0.05	0.61	1.05	-0.41
WAT	Healthcare	mid	0.98	0.45	0.7	0.51	0.38	0.28	-0.02	0.56	-0.01	0.84	0.88	0.84	1.12
KEYS	Technology	high	0.98	0.57	0.61	0.43	0.38	0.84	-0.01	1.42	-0.01	1.85	0.97	0.78	-0.62
WELL	Real Estate	low	0.98	0.61	0.44	0.85	0.5	0.39	-0.12	0.39	-0.04	-0.02	0.68	0.83	-0.71
MET	Financial	mid	0.98	0.51	0.68	0.26	0.36	0.25	-0.04	0.46	-0.03	0.66	0.76	0.89	0.77
HPE	Technology	high	0.98	0.63	0.6	0.72	0.46	1.26	-0.03	2.49	-0.01	2.05	1.6	0.96	-0.23
ABBV	Healthcare	low	0.97	0.53	0.43	0.61	0.42	0.13	-0.01	0.4	-0.05	-0.11	0.66	0.71	0.32
ILMN	Healthcare	mid	0.97	0.52	0.68	0.1	0.32	1.23	-0.01	2.34	0	0.8	1.32	2.09	-1.82
PANW	Technology	high	0.97	0.53	0.59	0.8	0.48	0.59	-0.01	1.91	0	1.18	1.26	0.81	-0.07
D	Utilities	low	0.96	0.63	0.43	0.62	0.46	0.13	-0.16	0.13	-0.04	-0.07	0.81	0.89	0.3
WBD	Communication Services	mid	0.96	0.47	0.68	0.91	0.52	0.47	-0	0.81	-0.04	0.5	1.88	2.23	1.92
AMD	Technology	high	0.96	0.53	0.57	0.74	0.4	1.69	-0.02	2.87	-0.01	3.2	0.79	0.61	0.48
KO	Consumer Defensive	low	0.96	0.52	0.42	0.01	0.22	0.37	-0.06	0.35	-0.03	-0.25	0.72	0.96	0.91
FTNT	Technology	high	0.96	0.46	0.57	0.14	0.36	0.81	-0	1.5	0	0.92	1	0.69	-0.6
PH	Industrials	mid	0.96	0.54	0.65	0.45	0.44	0.28	-0.11	0.38	-0.04	0.83	0.73	0.82	0.68
PPL	Utilities	low	0.95	0.58	0.42	0.86	0.5	-0.01	-0.17	0.04	-0.11	-0.03	0.78	0.89	-1.32
A	Healthcare	mid	0.95	0.47	0.64	0.84	0.48	0.09	-0.03	0.58	-0.01	0.85	1.2	1.14	-1.1
DELL	Technology	high	0.95	0.62	0.56	0.53	0.4	2.55	-0.07	4.03	-0.02	2.23	1.1	1.02	0.55

ticker	sector	beta_bucket	final_score	ml_score	comp_score	analog_pct	analog_hit_rate	mom_12_1	dist_52w_high	dist_52w_low	rs_lead_126	beta_252	rv20_pct_252	vol_dry_20_250	obv_price_div_63
PG	Consumer Defensive	low	0.95	0.5	0.41	1	0.66	-0.01	-0.11	0.08	-0.08	-0.03	0.9	0.84	1.41
CRL	Healthcare	high	0.95	0.54	0.55	0.57	0.46	0.67	-0	1.13	0	1.25	0.95	1.07	-1.08
VEEV	Healthcare	mid	0.95	0.45	0.63	0.83	0.46	-0.05	-0.09	0.91	-0.01	0.56	0.74	0.71	0.13

7. What to trust, what not to

Direct read of the evidence, in order of reliability.

1. What has out-of-sample support for 21-63d forward Sharpe in this universe

- Relative-strength leadership: `rs_lead_126` (RS line closer to its 6-month high than price is to its own). Walk-forward baseline, recent window: 42d IC 0.068 (NW t 2.4), 63d IC 0.082 (t 2.7); still 0.034-0.041 (t 2.1-2.2) after removing beta. The single most useful reading in the library.
- 12-1 momentum: recent beta-neutral 42d IC 0.048 (t 2.4), sector-excess IC 0.043 (t 2.1). Dead over 2014-2021 (full-window IC 0.005), alive since 2022-10. Consistent with the regime shift the mandate describes.
- Beta-bucketed walk-forward composite: recent 42d IC 0.047 (t 1.5), 63d 0.065 (t 1.9), top-minus-bottom decile 42d Sharpe spread +0.29 annualised with near-monotone deciles. The pooled composite has nothing (recent IC negative). Conditioning on beta is what makes the composite work. Its beta-neutral IC is small (0.006-0.015): inside each bucket it still leans on correlation / up-capture type factors, so part of its recent edge is bull-regime beta.
- Low-beta / defensive names respond to volume structure, not price structure: OBV leading price (`obv_price_div_63` , t 3.6 in defensives), volume dry-up vs 250d (`vol_dry_20_250` , t -2.1 in the low-vol tercile), falling dollar volume (`dvol_trend_21_126` , t -2.9 in low beta). Treat as a cluster, not three independent facts.

2. What did not show signal at this horizon in S&P 500 names (tested, not assumed)

- Compression setups (daily-range percentile, Bollinger width, NR7 clusters, base tightness), pooled: IC within +/-0.015 in every window and horizon. They only appear inside the low-vol / defensive bucket. The "compression then expansion" sequence factor is flat everywhere.
- Ignition-day volume multiple, volume z-score, big-volume up-day counts, the ignition x RS interaction: flat. Nothing in a 1-10 day volume burst predicts the next 1-3 months of risk-adjusted return here.
- Accumulation/distribution day counts, up/down volume ratio: flat pooled, mildly negative in places.
- Higher-low cadence, close location value, gap frequency, streaks, 20d range position: flat.

- The 100%-off-low "ceiling": wrong sign pooled and in high beta (names already 2x off the low kept producing better Sharpe, t +2.3 to +2.7); right sign only in Consumer Defensive (t -3.2). Do not use it as a universal exclusion.
- The LightGBM case-study model, in all five configurations: recent-window 42d IC between -0.001 and +0.023, none significant, every variant below the two single-factor baselines. A great 2019 (IC 0.11-0.14) and nothing stable since. Non-linear interaction mining over 86 factors does not beat two robust factors at this horizon with ~10 years of 500-name data. The analog columns in the ranking are therefore descriptive evidence per name, not a forecast.

3. Statistical honesty

- No single factor passes a 5% false-discovery-rate test across the 86-factor zoo in either window. The recent window has roughly 24 independent 42-day periods; $|t|$ of 2-2.7 is the ceiling a true IC of 0.05-0.08 can reach there. The claims above rest on (a) coherent clusters pointing the same way, (b) survival after beta-neutralisation, (c) walk-forward reproduction, not on any one t -stat.
- Multiple-testing in the conditional tables: 860 bucket tests (expected max $|t|$ under the null about 3.2) and 946 per-sector tests. Cells near 3 are suggestive only.
- Survivorship: today's constituents. Ranks are robust to it; absolute forward-Sharpe levels (the deciles all sit near -1.1 in the recent window because SPY's own realised Sharpe was high) are not the point.
- Variant selection (5 ML configurations, 2 composite variants) is a mild in-sample choice and is disclosed; the picked composite is the beta-bucketed one.

4. How to use the ranking

- `final_score` is the within-beta-bucket composite percentile, so the top of the list mixes low-, mid- and high-beta leaders by design. Read it with `beta_bucket` : high-beta names are there for trend persistence (RS leading, 12-1 momentum, proximity to highs, recent 2x-off-low crossings); low-beta names are there for quiet volume-led accumulation.
- Expected magnitude: a top-decile vs bottom-decile difference of roughly 0.3 annualised Sharpe over the next 42 days, with the top decile beating the bottom in about 57% of windows. That is a real but modest edge; it is a screen that concentrates probability, not a signal that resolves individual names.
- Refit cadence: the factors turn over slowly (21d rank autocorrelation 0.7-0.9 for the ones that matter), so re-scoring weekly is sufficient; the walk-forward re-estimates weights annually and that was enough.

8. Files

file	content
factor_ic_summary.csv / factor_ic_detail.csv	single-factor IC, NW t, FDR flags, decile spreads, autocorrelation, all horizons / benchmarks / windows
factor_ic42_by_year.csv	yearly mean IC (42d, SPY-excess) per factor
conditional_ic_{21,42,63}.csv	IC by beta tercile, vol tercile, sector group
conditional_ic_42_per_sector.csv	IC per GICS-style sector
sector_neutral_ic.csv	IC of sector-demeaned ranks vs sector- and SPY-excess Sharpe
factor_rank_corr_recent.csv	factor rank-correlation matrix (recent window)
ml_oos_evaluation.csv , ml_variants_oos_evaluation.csv , ml_oos_ic42_by_year.csv	walk-forward model results
composite_oos_evaluation.csv , composite_weights_by_fold.csv	walk-forward composite results and the factors it chose each year
composite_final_weights.csv , ml_final_feature_importance.csv	what the live score is built from
current_rankings.csv	the live ranking with analog evidence

9. Reproduce

```
pip install -e . scikit-learn lightgbm pyarrow scipy statsmodels tabulate
python -m pvv_score.data_io          # download
python -m pvv_score.build            # features + targets
python -m pvv_score.run_eval         # single-factor
python -m pvv_score.run_conditional # conditional
python -m pvv_score.model            # walk-forward ML
python -m pvv_score.run_model_variants
python -m pvv_score.composite        # walk-forward composite
python -m pvv_score.score_now        # live ranking
python -m pvv_score.report
```